

6 The matrix $\mathbf{M}$ is $\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$.

(a) Calculate $\mathbf{M}^2$, $\mathbf{M}^3$ and $\mathbf{M}^4$. **[2]**

(b) Hence make a conjecture about the matrix $\mathbf{M}^n$. **[1]**

(c) Prove your conjecture. **[5]**